Emergence of the Introduced Grass *Agropyron cristatum* and the Native Grass *Bouteloua gracilis* in a Mixed-grass Prairie Restoration

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Abstract

Prairie restoration at the northern edge of the Great Plains can be frustrated by previously established non-native perennial grasses. We compared the emergence of a widely introduced grass, *Agropyron cristatum*, and a common native grass, *Bouteloua gracilis*, in a 4-year-old field experiment in which the *Agropyron*-dominated vegetation had either been left intact or treated annually with herbicide. This was done at two levels of water supply, reflecting conditions expected in wet and dry years, to examine the effects of among-year variability in precipitation. Water addition significantly increased the emergence of both surface-sown and buried (1 cm deep) seeds. Herbicide treatment of neighbors did not increase the emergence of experimentally added seeds. Emergence was much greater for buried (80%) than surface-sown seeds (20%). Significantly more *Bouteloua* than *Agropyron* germinated from experimentally buried seeds. Whereas only a single seedling of *Bouteloua* emerged from the existing seed bank, the mean density of *Agropyron* seedlings emerging from the seed bank was 930/m^2 (range, 0 to 6,455/m^2). Surprisingly, the emergence of *Agropyron* from the seed bank was not decreased by 4 years of herbicide treatment, possibly because herbicide may release *Agropyron* from intraspecific competition and allow increased seed production to compensate for decreased plant abundance. In summary, we found few differences between *Agropyron* and *Bouteloua* in spring and summer emergence at high or low water availability. The persistence of *Agropyron* stands despite repeated herbicide application may be partly due to increased seed production.

Key words: ***Agropyron cristatum*, *Bouteloua gracilis*, emergence, grass, grassland, Great Plains, herbicide, introduced grass, native grass, neighbor control, seed bank, water availability.**

Introduction

One of the greatest challenges to prairie restoration on the northern edge of the Great Plains is the persistence of introduced grasses (Wilson 2002), such as *Agropyron cristatum* (L.) Gaertn. (crested wheatgrass). *Agropyron* has been planted on 6 to 10 million ha in the northern Great Plains (Lesica & DeLuca 1996). Stands of *Agropyron* are characterized by low diversity, low root mass, low inputs of C and N to soils (Christian & Wilson 1999), and the ability to spread into native grassland (Heidinga & Wilson 2002). Control of *Agropyron* increases the establishment of native species (Wilson & Gerry 1995; Bakker et al. in press), but the large seed bank (Marlette & Anderson 1986; Coffin & Lauenroth 1989) and high growth rates (Gerry & Wilson 1995) of *Agropyron* relative to native grasses allow it to regrow after herbicide application and to form very persistent stands (Looman & Heinrichs 1973; Christian & Wilson 1999). For example, treating stands of *Agropyron* with a high rate of the herbicide glyphosate (1.1 kg/ha) in the spring reduced *Agropyron* biomass by only a third by the following August (Bakker at al. 1997). Two causes of persistence are vegetative growth from basal meristems and emergence from the seed bank. Here we examine seedling emergence of *Agropyron* and of a common native grass, *Bouteloua gracilis* (HBK) Lag. (blue grama), in an old field dominated by *Agropyron.*

Two factors are of paramount importance to prairie grass emergence: neighbors and water (Baskin & Baskin 1998). Neighbors are relatively easy to reduce using herbicides (Grossbard & Atkinson 1985), and this typically increases grass establishment (Coffin & Lauenroth 1988; Fowler 1988; Reader 1993; Bakker et al. in press). The large among-year variation in precipitation on the Great Plains (Briggs & Knapp 1995) suggests that the outcomes of any restoration effort may vary among years with different amounts of precipitation. We expected *Agropyron* and *Bouteloua* to differ in their responses to water for three reasons. First, *Bouteloua* is a C_4 plant and thus might be less susceptible to drought. In contrast, the adventitious roots of *Bouteloua* arise close to the soil surface relative to those of *Agropyron*, which may make it more susceptible to drought during the initial growth of adventitious roots (Hyder et al. 1971). Finally, the small seed size of *Bouteloua* (0.4 mg, Ambrose, unpublished data) relative to

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Agropyron (3.1 mg) may also make young seedlings relatively susceptible to drying (Leishman & Westoby 1994) and competition from neighbors (Turnbull et al. 1999). Standardized germination tests provide no information about differences in species responses to water because water is supplied for these tests. Both *Agropyron* and *Bouteloua* germinate in the range of 20 to 30°C; only *Agropyron* requires a cold pretreatment (Association of Official Seed Analysts 1988).

Our objectives were to examine the field emergence of *Agropyron* and *Bouteloua* under conditions of varying water supply and neighbor control. We examined emergence from three compartments: the seed bank, seeds broadcast on the soil surface, and seeds that were experimentally buried. We measured the seed bank to determine what proportion of seedlings originated from experimentally sown seeds and what proportion emerged from the seed bank. This also allowed us to document emergence from the seed bank with regards to composition and response to neighbors and water. We examined seeds sown on the surface because broadcasting seeds can be an effective method for establishing native grasses (Bakker et al. in press). We examined buried seeds to determine the fate of seeds in the soil.

Methods

Study Site

We worked in Grasslands National Park (49°22′N, 107°53′W) in southwestern Saskatchewan, Canada. The study site had been cultivated until 1946 or 1947 (Bakker et al. in press), at which time it was sown to *Agropyron cristatum*. *Agropyron cristatum*, *Selaginella densa* (spike moss), lichens, bare ground, and litter represented 81% of the total cover (Bakker et al. 1997). The surrounding vegetation was undisturbed mixed-grass prairie dominated by *Bouteloua gracilis*, *Stipa comata* (needle-and-thread grass), and *Selaginella densa* (Christian & Wilson 1999). Nomenclature follows Looman & Best (1987).

The climate was semiarid with mean daily temperatures of –13°C in January and 19°C in July, with about 155 snow-free days. Average annual precipitation is 313 mm (Environment Canada 1993) with most of the precipitation falling as rain during the growing season. The soil was brown clay loam (Agriculture Canada 1992).

Experimental Design

Five replicate plots (3 × 10 meters, 1 meter apart) were randomly assigned to each of two treatments: neighbors present or neighbors treated with the herbicide glyphosate (Round-up, Monsanto, St. Louis, MO, U.S.A.). In May 1994 glyphosate was sprayed at a rate of 1.1 kg/ha. In May 1995 to 1997 glyphosate was applied with a wick at a concentration of 2:1 water-to-glyphosate. We switched from spraying to wicking because wicking is unaffected by the windy conditions of the study area. The plots were part of a larger restoration experiment (Bakker et al. in press).

In April 1997, 10 tubes (polyvinyl chloride, 10 cm diameter, 15 cm long) were placed in each plot to allow water to be manipulated. Tubes were pounded into the ground to a depth of 11 cm. The top edges of tubes were left 4 cm above the soil to prevent seeds from blowing away and to prevent added water from flowing out. Tubes were set up in two rows of five tubes each with 1.3 meters between the rows and 0.5 meters between the tubes in each row.

Five tubes in each plot were randomly assigned to each of two watering levels (high or low). The timing and amount of water added was based on rainfall records from Val Marie, about 20 km southwest of the experiment. The high water level was patterned after the wettest year of the previous 20 years, 1975, in which total precipitation was 453.5 mm. The low water level was patterned after the driest year (1984, 194.0 mm). Deionized water was added weekly when appropriate (Fig. 1a). Natural precipitation was eliminated by covering each tube with a 20 × 20 × 0.3–cm sheet of transparent acrylic sheeting (97% transmittance) mounted horizontally 10 cm above the tubes. The five tubes for each watering level were used for studying soil moisture (two tubes) and seed emergence from broadcast seeds, the seed bank, and buried seeds (one tube each).

In summary, there were five replicate plots of each of two herbicide treatments, for a total of 10 plots. Each plot contained 10 tubes, 5 representing each of two watering levels. For each watering level two tubes measured soil moisture and three others measured three aspects of emergence (from broadcast seeds, the seed bank, and buried seeds).

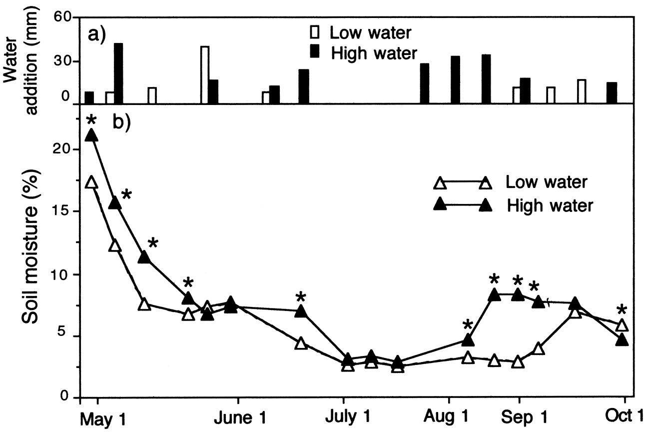

Figure 1. (a) The amount and timing of water addition at low (open bars) and high (solid bars) levels of water availability. The levels reflected the driest year and wettest year of the previous 20 years. (b) Soil moisture at the low (open triangle) and high (solid triangle) levels of water availability. Means were calculated across both levels of the herbicide treatment, which had no significant effect on soil moisture. Asterisks indicate significant ($p < 0.05$) differences between water levels.

Soil Moisture and Temperature

Soil moisture was measured gravimetrically. Four tubes (two from each water level) served as lysimeters in each plot. Immediately after installation these tubes were removed with soil enclosed. Fine mesh (1.5 × 1.5 mm) was taped to the bottom of each tube to hold the soil in place. The lysimeter was replaced in the hole. Lysimeters had the same precipitation-excluding covers as the other tubes. Water was added to the lysimeters at the same time as water was added to the other tubes. Lysimeters were weighed weekly using a portable balance. At the end of the season (25 October 1997) lysimeters were removed, dried to constant weight at 75°C, and weighed. Soil moisture for each week was calculated as % soil moisture = (WS – DS)/WS × 100, where WS is the mass of the wet soil and DS is the mass of the dry soil. We calculated the mean soil moisture of each water level treatment in each plot.

Soil temperature was measured weekly using a digital thermometer inserted 1 cm into the soil. Three measurements were taken in each tube and pooled.

Emergence

Seeds were obtained from commercial sources (*Agropyron*, United Grain Growers, Regina, Canada; *Bouteloua*, Enviroscapes, Lethbridge, Canada). Each seed was examined by gently squeezing and was used only if it was filled. Germination trials (Association of Official Seed Analysts 1988) using 400 filled seeds of each species determined viability (*Agropyron*, 67%; *Bouteloua*, 63%) and all numbers used hereafter refer to live seeds.

To examine the effect of herbicide and water on the emergence of seeds on the soil surface, we broadcast 200 seeds of each species into each of two tubes (one for each water level) in each plot on 25 April 1997. Vegetation was gently combed to ensure that seeds fell to the soil. Seedlings were counted and removed every 6 days until late June and every 7 to 10 days after that until 6 October.

Seeds emerging in the tubes could arise from two sources, either from broadcast seeds or from the seed bank. We found the number emerging from broadcast seeds by examining emergence from the seed bank in two tubes in each plot (one for each water level) to which no additional seeds were added. These tubes were examined for emergence at the same time as the tubes containing surface-sown seeds. The number of broadcast seeds of each species that emerged in each plot was found by subtracting the number of emergents from the seed bank (tubes with no seeds added) from the number of emergents found in tubes with seeds broadcast. In a few cases emergence from the seed bank exceeded the emergence of broadcast seeds, producing negative values for establishment in plots with broadcast seeds.

To examine the fate of buried seeds, we placed 200 seeds of each species in a bag (1 × 5 cm, sewed from nylon stockings) 1 cm below the soil surface of two tubes (one for each water level) in each plot on 25 April 1997. The bags allowed us to retrieve seeds at the end of the experiment. Tubes were checked weekly for emergence from the bags, and seedlings were counted and then pinched off at ground level to avoid seedling competition. Bags were removed on 25 October 1997. Seeds were stored dry in the refrigerator until they were sorted into field-germinated, dead (soft and not germinated), and ungerminated but potentially viable. Ungerminated seeds were tested for germinability (Association of Official Seed Analysts 1988), and those that germinated were recorded in a fourth class called "live."

Analysis

Data were tested for normality and heteroscedasticity and transformed as necessary. A three-factor split-split-plot analysis of variance tested the effects of herbicide, water, and species on the proportion of experimentally added seeds that emerged by the end of the experiment. The main plot effect was herbicide, the split-plot effect was water, and the split-split plot effect was species. A similar analysis was performed on species emerging from the seed bank, but species was not included as an effect in this analysis because only *Agropyron* emerged in sufficient numbers. Split-plot analysis of variance tested the effects of herbicide and water on soil moisture and soil temperature on each date, with herbicide as the main plot effect and water as the split-plot effect.

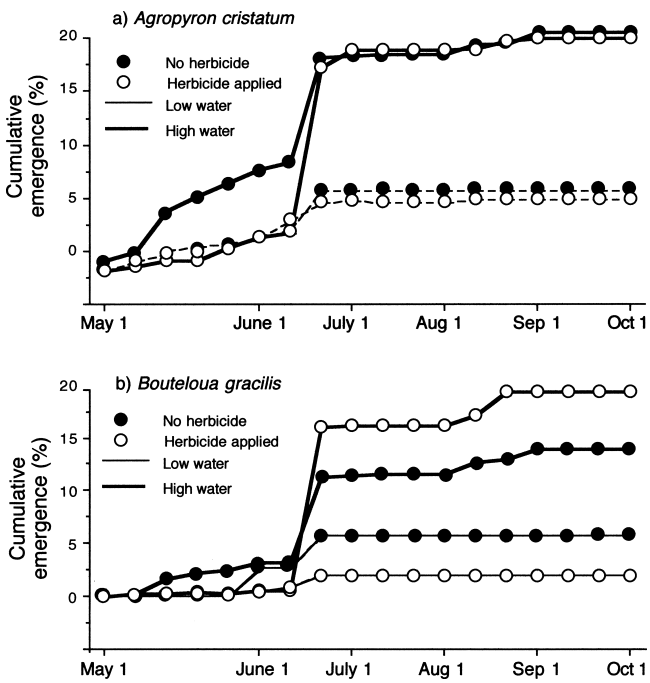

Figure 2. Cumulative emergence of surface-sown seeds of *Agropyron cristatum* (a) and *Bouteloua gracilis* (b) in tubes with and without herbicide and at low or high water availability. Final total emergence was significantly higher at the high level of water availability but did not vary significantly between herbicide treatments or between species.

Results

Soil moisture in the spring and fall was significantly higher in the high water treatment than in the low (Fig. 1b) but tended not to vary between water treatments during the summer. Soil temperature ranged between 15 and 45°C but did not vary significantly with water or herbicide treatment (data not shown).

Only one *Bouteloua* seedling emerged from the seed bank, but seedlings of *Agropyron* emerged from most (17/20) tubes, with an average density of 930/m^2 (range, 0 to 6,455/m^2). Emergence of *Agropyron* from the seed bank did not vary significantly with water or herbicide.

The emergence of surface-sown seeds was significantly increased by high water availability ($F_{1,16} = 20.6$, $p < 0.01$) but did not differ between *Agropyron* (Fig. 2a) and *Bouteloua* (Fig. 2b). Herbicide had no significant effect on emergence. Emergence of *Agropyron* from the seed bank exceeded emergence from sown seeds in some plots, causing an apparent negative average emergence of sown *Agropyron* in the early part of the experiment (Fig. 2a).

The cumulative emergence of both species was relatively low until early June (Fig. 2), although *Agropyron* germinated steadily in the high water treatment from mid-May to mid-June. For both species there was a substantial increase in emergence in the high water treatment in mid-June. No further emergence occurred in the low water treatment. In the high water treatment a few more seeds of both species germinated in July and August.

Emergence of buried seeds was significantly higher for *Bouteloua* than *Agropyron* (Fig. 3; $F_{1,16} = 136.3$, $p < 0.001$). As in the case of surface seeds, the emergence of buried seeds increased significantly with added water ($F_{1,16} = 14.2$, $p < 0.01$) but was unaffected by herbicide. The lack of a significant interaction between species and water treatments suggests that the emergence of buried seeds of both species was increased by water to about the same extent. Most (82% across all treatments) buried seeds emerged by the end of the experiment (Fig. 3). A few (3%) buried seeds were dead and had begun to decompose by the end of the experiment. There was no effect of herbicide, water, or species on the proportion of seeds that died. Of the buried seeds that were ungerminated and firm and potentially viable (14%), only 0.7% germinated in the growth chamber at the end of the experiment.

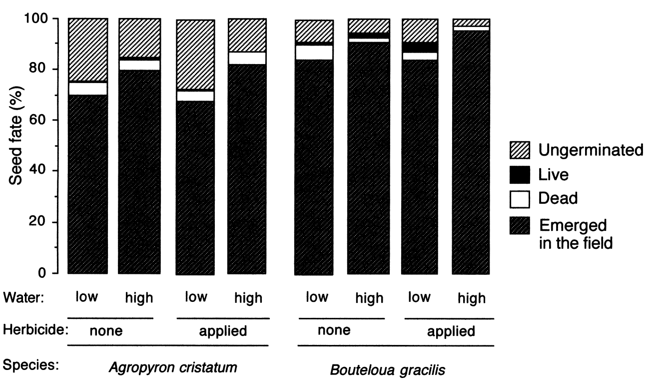

Figure 3. Fate of buried seeds of *Agropyron cristatum* (left) and *Bouteloua gracilis* (right) in tubes with and without herbicide and at low or high water availability. Final total emergence was significantly higher at the high level of water availability and was significantly higher for *Bouteloua* than *Agropyron*. "Live" refers to retrieved seeds which germinated in a growth chamber at the end of the growing season.

Discussion

Soil moisture varied significantly between water levels but not between herbicide treatments. Herbicide application might have been expected to increase soil moisture by reducing transpiration, given that herbicide application to these plots reduced shoot mass by one-third (Bakker et al. 1997). The results suggest that soil moisture is more influenced by abiotic factors, such as the timing and amount of precipitation, than by biotic factors, such as transpiration.

Emergence of both surface-sown and buried seeds was significantly higher at high water availability. There was, however, no significant difference between *Agropyron* and *Bouteloua* in their response (number of emergents from viable seeds) to water, as shown by the lack of a significant interaction term for the water and species effects. This is surprising because *Bouteloua* might be more sensitive to water availability because of its small size (Leishman & Westoby 1994) and relatively shallow adventitious roots (Hyder et al. 1971). In addition, the seasonal dynamics of emergence were almost identical between species. The C_3 *A. cristatum* tends to grow earlier in the season than the C_4 *B. gracilis*, and this difference was reflected to some extent by early emergence of *Agropyron* in the high water treatment. The fact that both species emerged at about the same time suggests that their field germination requirements may be more similar than the seasonal growth patterns of established plants.

Emergence of *Agropyron* from the seed bank was unaffected by water addition, whereas the emergence of experimentally added seeds was significantly increased by added water for both surface-sown seeds and buried seeds. The lack of a water effect on emergence from the seed bank might be attributable to the low density of seedlings emerging (930/m^2 averaged across other treatments) relative to the density that was experimentally sown (200 per tube = 25,465/m^2), making seed bank responses harder to detect. The considerable spatial variability of emergence from the seed bank (see Results) could be addressed by using more replicates.

Herbicide had no significant effect on the emergence of seeds, whether sown on the surface or buried. This is consistent with the lack of effect of herbicide on soil moisture; germination requires water, but soil moisture was unaffected by herbicide, so neighbor removal also had no effect on emergence. Similarly, herbicide did not increase emergence of *Bouteloua* planted with a seed drill at the site in an earlier experiment (Bakker et al. 1997). In contrast, herbicide significantly increased the establishment of surface-sown seeds at this site and of both surface-sown and drilled seeds at other sites in Saskatchewan (Wilson & Gerry 1995; Bakker et al. 1997). Similarly, the exclusion of neighbor roots from small plots significantly increased emergence of *Bouteloua* in Colorado (Aguilera & Lauenroth 1993). In summary, the effects of neighbor removal on emergence are not consistent (Fowler 1990), although significant effects are always positive.

Surprisingly, emergence from the seed bank of *Agropyron* was not decreased by 4 years of herbicide treatment. This presumably reflects a tripling in the density of seed heads of *Agropyron* in herbicide plots relative to controls (Wilson & Pärtel, unpublished data). Thus, an unexpected result of herbicide application may be to release *Agropyron* from intraspecific competition and to maintain its seed production and presence in the seed bank.

In addition to differences in their representation as seedlings emerging from the seed bank, *Agropyron* and *Bouteloua* also differed in their emergence from experimentally buried seeds. Emergence was significantly higher for *Bouteloua* than *Agropyron*, despite using the same number of viable seeds. This may be another factor contributing to the absence of *Bouteloua* from seed banks (Coffin & Lauenroth 1989) and the presence of *Agropyron* (Marlette & Anderson 1986). Higher emergence for the small-seeded *Bouteloua* contradicts the positive association between seed size and establishment found in a study of seven English grassland species (Turnbull et al. 1999).

The maximum final emergence was about 20% for surface-sown seeds but 80% for buried seeds. The opposite pattern was found in a greenhouse experiment, where emergence of *Bouteloua* fell with increasing depth over the range of 1.5 to 6 cm (Redmann & Qi 1992). The difference is probably attributable to the discontinuous water supply in our field experiment. Wetting and drying of seeds decreases germination rates in growth chambers (Doucet & Cavers 1996). In a greenhouse, with a steady water supply, emergence is negatively influenced by burial depth. In the field, however, burial may buffer seedlings from temporal variability in water availability (Hyder et al. 1971).

In summary, the significant effect of water availability on emergence suggests that natural among-year variation in precipitation will have a large influence on the success of restoration efforts. The maintenance of a seed bank of *Agropyron* in plots treated with herbicide creates an additional challenge for restoration.

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